ABSTRACT DOMAINS AND CUT POINTS IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES: JOINS OF CLOSURE OPERATORS ON A LATTICE AND OF CUT-POINT TOPOLOGIES ON A CONTROL-FLOW GRAPH

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Abstract. An abstract domain of Cousot and Cousot, a closure operator ρ on a complete lattice, determines the class W_ρ of pairs $x \le y$ that ρ does not distinguish, an element of the lattice of two-out-of-three classes. We show that $W_\rho \cap W_{\rho'}$ is the class of the reduced product, that $W_\rho \vee W_{\rho'}$ is the class of pairs with $y \le (\rho\rho')^k x$ for some k, and that it is the class of the least common abstraction iff the alternation $\rho\rho'\rho\rho'\cdots$ stops at a finite stage at every point; cancellation is needed when the closures do not preserve meets, as for reachability closures. On the presheaf topos of the path category of a directed graph, a set S of vertices gives the topology whose sheaves are the presheaves determined by their values at S. For two such sets the join of the two classes inverted by sheafification is the class of the join topology iff the alternations between S and T along paths avoiding $S \cap T$ are bounded; for a finite graph, iff no strongly connected component of the graph with $S \cap T$ removed meets both S and T, a linear-time test, and then the join topology is that of $S \cap T$.

1. Introduction

In one sentence: for two abstractions of the same concrete lattice, the join of the two classes of "pairs the abstraction cannot tell apart" is generated by alternating the two abstractions and cancelling, and it is the class of the least common abstraction exactly when the alternation is finite at every point; on the path category of a control-flow graph, with abstractions given by sets of cut points, the finiteness is a condition on closed walks, tested in linear time.

Let $\mathcal{C}$ be a category. A *sameness* on $\mathcal{C}$ is a class of morphisms containing the identities and closed under composition and two-out-of-three;

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samenesses form a complete lattice $\mathrm{Same}(\mathcal{C})$, the meet being the intersection and the join the least sameness containing both [10]. Every localisation of $\mathcal{C}$ contributes the class W of maps it inverts, and the question studied in [11, 12, 13, 14] is whether W sends the join of two localisations to the join of the two classes. For localisations given by closure operators on subobjects (Lawvere–Tierney topologies, hereditary torsion theories, nuclei) the answer of [14] is a finite-iteration criterion: in a Grothendieck topos or a Grothendieck abelian category, $W_c \vee W_{c'} = W_{\bar{c}}$ for the least closure operator $\bar{c}$ above c and c' iff $(cc')^k = \bar{c}$ for some k.

This paper applies the criterion in two settings suggested by the semantics of programs. The first is the lattice of abstract domains of abstract interpretation [2]: an abstract domain on a complete lattice L of concrete properties is an upper closure operator ρ on L, the closure operators form a complete lattice $\mathrm{uco}(L)$ [16], the reduced product of two domains is their meet and their least common abstraction is their join [5]. The category is the poset L itself, so that the maps are the inequalities $x \leq y$, and the class W_ρ consists of the inequalities that ρ collapses, $\rho x = \rho y$: the refinements y of x that the analysis cannot see. General abstract domains are not nuclei, they need not preserve binary meets (intervals, octagons, reachability under a transition relation), and the criterion of [14] for nuclei on a frame does not apply to them as it stands. Theorem 2.3 treats arbitrary closure operators on a complete lattice: the meet is always preserved, the join is the class of pairs $x \leq y$ with $y \leq (\rho\rho')^k x$ for some k, it is the class of $\rho \sqcup \rho'$ iff the alternation reaches $\rho \sqcup \rho'$ at a finite stage at every x, and, unlike the case of nuclei, the two-out-of-three cancellation is needed to generate the join from the alternating chains (already on the four-element Boolean lattice). Section 3 spells this out for the reachability closures $S \mapsto R^*(S \cup I)$ of transition relations: the join of the two classes is bounded alternation of the two relations, it is the class of the union relation iff the alternation depth is finite from every set of states, which holds for finite state spaces with bound the number of states and fails on $\mathbb{N}$ for the even and the odd steps.

The second setting is the presheaf topos on the path category of a directed graph G, a control-flow graph: a presheaf is a flow of sets along the control flow, and a set S of vertices, a set of cut points in the sense of Bourdoncle [1], determines the Grothendieck topology J_S whose sheaves are the presheaves determined by their values at S, $\mathrm{Sh}(J_S) \simeq \mathrm{Psh}(S)$. Theorem 4.3 says that $W_{J_S} \vee W_{J_T} = W_{J_S \vee J_T}$ iff the number of alternations between visits to T and visits to S along paths that avoid $S \cap T$ is bounded; for a finite graph, iff no closed walk avoiding $S \cap T$ meets both S and T, i.e. no strongly connected component of G with $S \cap T$ removed meets both, a condition decided in linear time; the bound is then $\min(|S \setminus T|, |T \setminus S|)$ and the join topology is $J_{S \cap T}$. The theorem interpolates between two known facts: on a finite category every pair of topologies has a preserved join [12], which is the acyclic case, and on the poset $\mathbb{N}$ the topologies of the even and of the odd numbers do not [11], which is the infinite path with unbounded alternation. A third section

records the finite action monoids: there the ideal topologies are indexed by sets of idempotent actions, resets, and the reading of the closure is "every future through a reset lands in the invariant".

What is known. Samenesses are from [10]; the finite-iteration criterion in a Grothendieck topos from [13], in exact categories with closure operators, including nuclei on a frame as the poset case, from [14]; the finite-category join theorem from [12]; the counterexample on $\mathbb{N}$ from [11]. Abstract domains as closure operators, the lattice of abstract domains, the reduced product and the least common abstraction are in [2, 16, 5, 6], and the computation of the join of two closure operators by transfinite alternation (Lemma 2.1) is standard and is recalled only to fix notation; the iterated reduction computing the reduced product on a product domain is in [7, 3]; cut sets of control-flow graphs in [1]. Lawvere–Tierney topologies, sheafification and the bidense description of the maps it inverts are in [9, 8].

What is new. Theorem 2.3 (closure operators on a complete lattice without meet preservation: meet, join as alternating chains plus one cancellation, criterion, necessity of cancellation), Proposition 3.1 (reachability closures) and Theorem 4.3 (cut-point topologies on the path category of a graph, with the closed-walk criterion, its linear-time test and the identification of the join topology), together with the dictionary of Section 5. The engine is the criterion of [14]; what is added is its application to closures that are not nuclei and to the object-generated topologies of infinite path categories, where the combinatorics of alternation along paths decides the question.

2. Closure operators on a complete lattice

Let L be a complete lattice, regarded as a category with a morphism $x \to y$ iff $x \leq y$. A *closure operator* on L is a map $\rho\colon L \to L$ that is extensive ($x \leq \rho x$), monotone and idempotent; its set of fixed points $\mathrm{Fix}\rho = \rho(L)$ is closed under arbitrary meets, and ρx is the least fixed point above x. The closure operators on L form a complete lattice $\mathrm{uco}(L)$ under the pointwise order [16], which is the reverse inclusion of fixed-point sets; the meet of ρ, ρ' is $x \mapsto \rho x \wedge \rho' x$, with fixed points the meets $a \wedge b$, $a \in \mathrm{Fix}\rho$, $b \in \mathrm{Fix}\rho'$, and the join $\rho \sqcup \rho'$ is the closure operator with fixed points $\mathrm{Fix}\rho \cap \mathrm{Fix}\rho'$. In abstract interpretation [2, 5], $\rho \leq \rho'$ means that ρ is more precise, the meet is the reduced product and the join the least common abstraction.

Lemma 2.1 (standard)**.** *For closure operators ρ, ρ' and $x \in L$ define $x_0 = x$, $x_{\alpha+1} = \rho\rho' x_\alpha$ and $x_\lambda = \bigvee_{\alpha<\lambda} x_\alpha$ at limits. The sequence is increasing, stabilises at some ordinal, and its value is $(\rho \sqcup \rho')x$.*

Proof. If $x_{\alpha+1} = x_\alpha =: z$ then $\rho' z \leq \rho\rho' z = z$ gives $\rho' z = z$ and then $\rho z = \rho\rho' z = z$, so $z \in \mathrm{Fix}\rho \cap \mathrm{Fix}\rho'$; every common fixed point above x is above all x_α, so z is the least one. □

Definition 2.2. For a closure operator ρ on L put

$$W_\rho := \{x \le y : \ \rho x = \rho y\} = \{x \le y : \ y \le \rho x\},$$

the inequalities collapsed by ρ, i.e. the refinements y of x that ρ does not distinguish from x.

W_ρ is a sameness on the poset L: it contains the identities, is closed under composition ($x \le y \le z$ with $z \le \rho y \le \rho\rho x = \rho x$), and under both cancellations ($x \le y \le z$ with $z \le \rho x$ gives $y \le \rho x$ and $z \le \rho x \le \rho y$). No meet preservation is used. If $\rho \le \rho'$ then $W_\rho \subseteq W_{\rho'}$: a less precise analysis collapses more.

Theorem 2.3. *Let ρ, ρ' be closure operators on a complete lattice L.*

(i) *$W_\rho \cap W_{\rho'} = W_{\rho \sqcap \rho'}$: the reduced product distinguishes $x \le y$ iff one of the two abstractions does.*

(ii) *$W_\rho \vee W_{\rho'} = \{x \le y : \ y \le (\rho\rho')^k x$ for some $k \ge 0\}$ in $\mathrm{Same}(L)$; for such a pair, with $z = (\rho\rho')^k x$, both $x \le z$ and $y \le z$ are alternating chains of steps in $W_{\rho'}$ and W_ρ, so every element of the join is obtained from two alternating chains by one cancellation.*

(iii) *$W_\rho \vee W_{\rho'} = W_{\rho \sqcup \rho'}$ iff for every $x \in L$ there is k with $(\rho\rho')^k x = (\rho \sqcup \rho')x$: the alternation of Lemma 2.1 stops at a finite stage at every point.*

(iv) *$\rho\rho' = \rho \sqcup \rho'$ (one alternation suffices uniformly) iff $\rho\rho'$ is idempotent iff ρ maps $\mathrm{Fix}\rho'$ into $\mathrm{Fix}\rho'$.*

(v) *The compositional closure of $W_\rho \cup W_{\rho'}$, the class of alternating chains, can be strictly smaller than the join; it equals the join when ρ and ρ' preserve binary meets.*

Proof. (i) $x \le y$ lies in $W_{\rho \sqcap \rho'}$ iff $y \le \rho x \wedge \rho' x$ iff $y \le \rho x$ and $y \le \rho' x$.

(ii) Let D be the right-hand side. D is a sameness: if $y \le (\rho\rho')^k x$ and $z \le (\rho\rho')^l y$ then $z \le (\rho\rho')^{k+l} x$; if $x \le y \le z$ and $z \le (\rho\rho')^k x$ then $y \le (\rho\rho')^k x$ and $z \le (\rho\rho')^k x \le (\rho\rho')^k y$. It contains W_ρ and $W_{\rho'}$ ($y \le \rho x \le \rho\rho' x$, $y \le \rho' x \le \rho\rho' x$). So the join J of W_ρ and $W_{\rho'}$ is contained in D. Conversely let $x \le y \le z := (\rho\rho')^k x$, and put $x_0 = x$, $x_{2i+1} = \rho' x_{2i}$, $x_{2i+2} = \rho x_{2i+1}$, so that $x_{2k} = z$. The chain $x = x_0 \le x_1 \le \cdots \le x_{2k} = z$ has its steps alternately in $W_{\rho'}$ and W_ρ, so $(x, z) \in J$. The chain $y = y \vee x_0 \le y \vee x_1 \le \cdots \le y \vee x_{2k} = z$ (the last equality as $y \le z$) has the same property: if $x_{j+1} = d x_j$ with $d \in \{\rho, \rho'\}$, then $y \vee x_{j+1} \le d(y \vee x_j)$, both y and $d x_j$ being below $d(y \vee x_j)$; so $(y, z) \in J$. Then $(x, y) \in J$ by cancellation from (x, z) and (y, z).

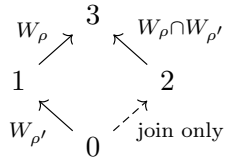
(iii) $W_{\rho \sqcup \rho'} = \{x \le y : y \le (\rho \sqcup \rho')x\}$ contains D, since $(\rho\rho')^k x \le (\rho \sqcup \rho')x$. If every fibre top $(\rho \sqcup \rho')x$ is some $(\rho\rho')^k x$ the two classes agree; if for some x no $(\rho\rho')^k x$ equals $(\rho \sqcup \rho')x$, then $(x, (\rho \sqcup \rho')x)$ lies in $W_{\rho \sqcup \rho'}$ but not in D, as $(\rho \sqcup \rho')x \le (\rho\rho')^k x$ would force equality.

(iv) If $\rho\rho'$ is idempotent and $a \in \mathrm{Fix}\rho'$, then $\rho' \rho a \le \rho\rho' \rho a = \rho\rho' \rho\rho' a = \rho\rho' a = \rho a$, so $\rho a \in \mathrm{Fix}\rho'$. If ρ preserves $\mathrm{Fix}\rho'$ then $\rho\rho' x \in \mathrm{Fix}\rho'$, so $\rho\rho' \rho\rho' x =$

$\rho\rho\rho' x = \rho\rho' x$. An idempotent $\rho\rho'$ is a closure operator above ρ and ρ' and below every closure operator above both, so it is $\rho \sqcup \rho'$; conversely $\rho \sqcup \rho'$ is idempotent.

(v) Example 2.4 gives a strict inclusion. If ρ, ρ' preserve binary meets, let $x \le y \le z = x_{2k}$ with the chain x_j of (ii) and put $u_j = y \wedge x_j$, so $u_0 = x$, $u_{2k} = y$ and $u_j \le u_{j+1}$. If $x_{j+1} = dx_j$ then $u_{j+1} = y \wedge dx_j \le dy \wedge dx_j = d(y \wedge x_j) = du_j$, so $u_j \le u_{j+1}$ lies in W_d and $x = u_0 \le \cdots \le u_{2k} = y$ is an alternating chain. (When L is a frame this is the poset case of Theorem A of [14].) □

Example 2.4. Let $L = \{0 < 1, 2 < 3\}$ be the four-element Boolean lattice, ρ the closure with $\mathrm{Fix}\rho = \{0, 3\}$ ($\rho 1 = \rho 2 = 3$) and ρ' the closure with $\mathrm{Fix}\rho' = \{1, 3\}$ ($\rho' 0 = 1$, $\rho' 2 = 3$). Then $W_\rho = \{1 \le 3, 2 \le 3\}$, $W_{\rho'} = \{0 \le 1, 2 \le 3\}$ (plus identities), $\rho\rho' 0 = 3$, so $0 \le 2$ lies in the join by (ii); but the only non-identity step out of 0 in $W_\rho \cup W_{\rho'}$ is $0 \le 1$, and $1 \not\le 2$, so $0 \le 2$ is not an alternating chain: it is obtained from $0 \le 3$ and $2 \le 3$ by cancellation. Here $\rho(1 \wedge 2) = 0 \ne 3 = \rho 1 \wedge \rho 2$.



Remark 2.5. In a finite lattice the alternation of Lemma 2.1 stops after at most $|L|$ steps, so (iii) always holds there and the content of the theorem is (ii) and (v): the join is generated by alternating chains *and* cancellation. On an infinite lattice (iii) can fail; the simplest instance is in Section 3.

3. Reachability closures

Let X be a set of states, $R \subseteq X \times X$ a transition relation and $I \subseteq X$ a set of initial states. On the complete lattice $L = \mathcal{P}(X)$ put

$$\rho_{R,I}(S) := R^*(S \cup I),$$

the set of states reachable from S or from I along R: the collecting semantics of "run R from S".

Proposition 3.1. (i) *$\rho_{R,I}$ is a closure operator on $\mathcal{P}(X)$ with fixed points the R-closed sets containing I (the inductive invariants); it preserves binary intersections only in degenerate cases.*

(ii) *$\rho_{R,I} \sqcap \rho_{R',I'}$ is $S \mapsto R^*(S \cup I) \cap R'^*(S \cup I')$ and $\rho_{R,I} \sqcup \rho_{R',I'} = \rho_{R \cup R', I \cup I'}$; for $k \ge 1$, $(\rho_{R,I}\rho_{R',I'})^k S$ is the set of states reachable from $S \cup I'$ by a path of at most k blocks, each an R'-path followed by an R-path, together with the states reachable from I by an R-path followed by at most $k-1$ such blocks (the initial states of the first system enter only after the R'-phase of a block).*

(iii) $W_{\rho_{R,I}} \vee W_{\rho_{R',I'}}$ *is the class of pairs* $S \subseteq T$ *with* $T \subseteq (\rho_{R,I}\rho_{R',I'})^k S$ *for some* k*, i.e. every state of* T *is reachable in the sense of (ii) with a bounded number of blocks, and it equals* $W_{\rho_{R\cup R',I\cup I'}}$ *iff the alternation depth of* $(R \cup R')$*-reachability is finite from every* S*. If* X *is finite this holds with at most* $|X|$ *blocks.*

(iv) *On* $X = \mathbb{N}$ *with* $R = \{(2i, 2i+1)\}$*,* $R' = \{(2i+1, 2i+2)\}$ *and* $I = I' = \emptyset$*,* $(\rho_R\rho_{R'})^k\{0\} = [0, 2k-1]$ *for* $k \geq 1$ *(the first* R'*-step from* 0 *is empty), which never reaches* $\rho_{R\cup R'}\{0\} = \mathbb{N}$*, so* $W_{\rho_R} \vee W_{\rho_{R'}} \subsetneq W_{\rho_{R\cup R'}}$*.*

Proof. (i) Extensive, monotone and idempotent since R^* is reflexive and transitive; the fixed points are the sets closed under R and containing I. For $x \xrightarrow{R} z \xleftarrow{R} y$ with $x \neq y$ one has $z \in \rho\{x\} \cap \rho\{y\}$ but $\rho(\{x\} \cap \{y\}) = \rho\emptyset = R^*I$. (ii) The meet is pointwise; the common fixed points of the two closures are the sets closed under R and R' containing $I \cup I'$, the fixed points of $\rho_{R\cup R',I\cup I'}$, which is therefore the join. For the iterates, $\rho_{R,I}\rho_{R',I'}(A) = R^*R'^*(A \cup I') \cup R^*I$; writing $B_k(A)$ for the states reachable from A by at most k blocks, so that $B_1B_k = B_{k+1}$ and $B_k(A) \subseteq B_{k+1}(A)$ as blocks may be empty, induction on k gives $(\rho_{R,I}\rho_{R',I'})^k S = B_k(S \cup I') \cup B_{k-1}(R^*I)$. (iii) is Theorem 2.3(ii), (iii); if X is finite, every block that changes the set adds a state, so the iteration of Lemma 2.1 stops after at most $|X|$ blocks. (iv) is a direct computation, and Theorem 2.3(iii). □

Remark 3.2. Example 2.4 is of this form: $X = \{p, q\}$, $R = \{(p, q), (q, p)\}$, $I = \emptyset$ and $R' = \emptyset$, $I' = \{p\}$. The pair $\emptyset \subseteq \{q\}$ is collapsed by the join (from $\emptyset$ the second system reaches p and the first then q) but by no alternating chain of collapses, the only refinement of $\emptyset$ collapsed by one of the two systems being $\{p\} \not\subseteq \{q\}$.

Remark 3.3. The iterated reduction of Granger [7] and of Cousot, Cousot and Mauborgne [3], which alternately refines the two components of a pair (a, b) in a product of abstract domains and is the Nelson–Oppen procedure when the domains are logical theories, computes the reduced product, the *meet* $\rho \sqcap \rho'$, by a decreasing iteration on the product domain. The alternation $\rho\rho'\rho\rho'\cdots$ of Lemma 2.1 computes the *join*, the least common abstraction, by an increasing iteration on L. The two are not to be confused: the first terminates in one step in the sense of (i) of Theorem 2.3, the second is governed by (iii).

4. Cut-point topologies on the path category of a graph

Let G be a directed graph, finite or infinite, with vertex set V, and let $\mathcal{P}(G)$ be its path category: objects the vertices, morphisms the finite paths, identities the empty paths. A presheaf X on $\mathcal{P}(G)$ is a set $X(c)$ for each vertex and a restriction $X(c) \to X(d)$, $x \mapsto x \cdot f$, for each path $f\colon d \to c$, a flow of sets along the control flow; a subpresheaf is a family of subsets closed under restriction. Write $\mathcal{E} = \mathrm{Psh}(\mathcal{P}(G))$, a Grothendieck topos.

For a set $S \subseteq V$ of vertices, cut points, let D_S be the set of paths that pass through a vertex of S, the empty paths at vertices of S included. D_S is a two-sided ideal of $\mathcal{P}(G)$ (closed under composition with arbitrary paths on either side) and idempotent, $D_S \circ D_S = D_S$, because $f = f \circ \mathrm{id}_s$ for a path f through $s \in S$. For a vertex c let $D_S(c)$ be the sieve of paths into c through S.

Lemma 4.1. *The sieves containing $D_S(c)$, for all c, form a Grothendieck topology J_S on $\mathcal{P}(G)$. Its sheaves are the presheaves X with $X \cong \mathrm{Ran}_i(X|_S)$ for the inclusion i of the full subcategory of $\mathcal{P}(G)$ on the vertices of S, so that restriction gives $\mathrm{Sh}(J_S) \simeq \mathrm{Psh}(S)$, where $\mathrm{Psh}(S)$ denotes the presheaves on that full subcategory (its morphisms are all paths of G between vertices of S, not only the edges inside S), and its closure operator on subpresheaves $A \subseteq X$ is*

$$c_S(A)(c) = \{x \in X(c) :\ x \cdot f \in A \text{ for every path } f\colon s \to c \text{ with } s \in S\}.$$

For two sets S, T one has $c_S c_T = c_{D_S \circ D_T}$, where $c_D(A)(c) = \{x : x \cdot f \in A\ \forall f \in D(c)\}$ for an ideal D; so $(c_S c_T)^k = c_{(D_S \circ D_T)^k}$, and $D_S \circ D_T$ consists of the paths with a visit to T followed by a visit to S.

Proof. Stability: the pullback of $D_S(c)$ along $g\colon e \to c$ is $\{h : gh \in D_S(c)\} \supseteq D_S(e)$. Transitivity: let R be a sieve on c with $h^*R \supseteq D_S(\mathrm{dom}\, h)$ for all $h \in D_S(c)$; for $f = u \circ v \in D_S(c)$ with $u\colon s \to c$, $s \in S$, we have $u \in D_S(c)$ and $\mathrm{id}_s \in D_S(s) \subseteq u^*R$, so $u \in R$ and $f \in R$. Restriction $\mathrm{Psh}(\mathcal{P}(G)) \to \mathrm{Psh}(S)$ is left exact with right adjoint Ran_i, and Ran_i is fully faithful since i is, so restriction is the inverse image of a geometric embedding; a sieve $R \hookrightarrow y(c)$ is inverted by restriction iff R contains every path from a vertex of S into c, i.e. iff $R \supseteq D_S(c)$, so the embedding is $\mathrm{Sh}(J_S)$ [8, A4.3, A4.5]. The closure of A for J_S consists of the x whose sieve $\{f : x \cdot f \in A\}$ is covering, which is the displayed formula; and $c_D c_E(A) = \{x : x \cdot f \in c_E(A)\ \forall f \in D\} = \{x : x \cdot f \cdot g \in A\ \forall f \in D, g \in E\} = c_{D \circ E}(A)$, with $f \circ g$ the path g followed by f. □

Lemma 4.2. *A path f lies in $(D_S \circ D_T)^k$ iff there are vertices $t_1, s_1, \ldots, t_k, s_k$ occurring along f in this order, positions non-decreasing, with $t_i \in T$ and $s_i \in S$. A path through a vertex of $S \cap T$ lies in $(D_S \circ D_T)^k$ for every k. Hence the chain $(D_S \circ D_T)^k$ stabilises iff the number of alternations "a visit to T, then a visit to S" along paths avoiding $S \cap T$ is bounded, and then $(D_S \circ D_T)^{a+1} = D_{S \cap T}$ for the bound a.*

Proof. $f \in (D_S \circ D_T)^k$ iff $f = u_k v_k \cdots u_1 v_1$ with $v_i \in D_T$, $u_i \in D_S$, the splitting points being arbitrary vertices along f and the factors possibly empty; this is the stated condition. At a common vertex x all factors may be taken empty. If the alternation along paths avoiding $S \cap T$ is bounded by a, a path in level $a+1$ passes through $S \cap T$, and conversely such paths are in every level; if it is unbounded, the chain drops at infinitely many levels. □

Theorem 4.3. *Let G be a directed graph and $S, T \subseteq V$. In* $\mathrm{Same}(\mathrm{Psh}(\mathcal{P}(G)))$*,*

$$W_{J_S} \vee W_{J_T} = W_{J_S \vee J_T}$$

iff the number of alternations between visits to T and visits to S along paths of G avoiding $S \cap T$ is bounded, and then $J_S \vee J_T = J_{S \cap T}$, $\mathrm{Sh}(J_S) \cap \mathrm{Sh}(J_T) \simeq \mathrm{Psh}(S \cap T)$*. If G is finite this holds iff no closed walk of G avoiding $S \cap T$ meets both S and T, i.e. iff no strongly connected component of the induced subgraph $G - (S \cap T)$ contains both a vertex of S and a vertex of T; the bound is then at most* $\min(|S \setminus T|, |T \setminus S|)$*, and the condition is decided in time $O(|V| + |E|)$ by a decomposition into strongly connected components. In particular, for disjoint S, T on a finite graph the join is preserved iff no strongly connected component of G meets both S and T.*

Proof. Let $c = c_S$, $c' = c_T$ and let $\bar{c}$ be the least closure operator above both, the closure operator of the join topology; by [14, Thm. B], $W_{J_S} \vee W_{J_T} = W_{\bar{c}}$ iff $(cc')^k = \bar{c}$ for some k. If the alternation is bounded by a, then by Lemmas 4.1 and 4.2 $(cc')^{a+1} = c_{D_{S \cap T}}$ is idempotent, hence a closure operator above c and c', and every closure operator above both is above cc' and so above $(cc')^{a+1}$; thus $\bar{c} = (cc')^{a+1} = c_{S \cap T}$, the criterion holds, and the join topology is $J_{S \cap T}$. If the alternation is unbounded, the chain of ideals $(D_S \circ D_T)^k$ does not stabilise by Lemma 4.2, and an ideal D is determined by c_D: for $X = y(c)$, $D(c)$ is the least sieve A on c with $\mathrm{id}_c \in c_D(A)(c)$. Since $c, c' \leq \bar{c}$ and $\bar{c}$ is idempotent, $(cc')^k \leq \bar{c}$ for all k and $c\bar{c} = c'\bar{c} = \bar{c}$. If $\bar{c} = (cc')^k$ for some k, then $(cc')^m = (cc')^{m-k}\bar{c} = \bar{c} = (cc')^k$ for all $m \geq k$, so $(D_S \circ D_T)^m = (D_S \circ D_T)^k$ for all $m \geq k$ and the chain stabilises, a contradiction; so the criterion fails. For finite G: if a closed walk avoiding $S \cap T$ meets $s \in S$ and $t \in T$, going round it m times gives alternation $\geq m$; conversely a path avoiding $S \cap T$ with alternation $\geq |S \setminus T| + 1$ repeats an S-vertex, $s_i = s_j$ with $i < j$, with a T-visit t_{i+1} strictly between (a visit to S and one to T at the same position would be a vertex of $S \cap T$), and the subwalk from s_i to s_j is a closed walk avoiding $S \cap T$ meeting S and T; likewise with $T \setminus S$ and the t_i. Two vertices lie on a common closed walk of $G - (S \cap T)$ iff they lie in the same strongly connected component, and the components are computed in linear time [15]. □

Example 4.4. (a) G: $s \to t \to s'$, $S = \{s, s'\}$, $T = \{t\}$. The alternation is at most one, the join is preserved, and $J_S \vee J_T = J_\emptyset$ is the degenerate topology: a J_S-sheaf has $X(t) \cong X(s)$, a J_T-sheaf has $X(s) = 1$ and $X(s') \cong X(t)$, so a common sheaf is terminal. (b) G: the two-cycle $s \rightleftarrows t$, $S = \{s\}$, $T = \{t\}$. Going round the cycle gives unbounded alternation, and $W_{J_S} \vee W_{J_T} \subsetneq W_{J_S \vee J_T}$; adding the vertex s to T restores preservation, as every closed walk then meets $S \cap T = \{s\}$. (b′) G: $s \rightleftarrows u \rightleftarrows t$, $S = \{s\}$, $T = \{t\}$. The simple cycles $s \to u \to s$ and $u \to t \to u$ each meet only one of the two sets, but the closed walk $s \to u \to t \to u \to s$ meets both, the alternation is unbounded and the join is not preserved: the condition of Theorem 4.3 is on closed walks, equivalently on strongly connected components, not on

simple cycles. (c) G the infinite path $0 \to 1 \to 2 \to \cdots$, S the even and T the odd vertices: no cycles, unbounded alternation, and the failure of [11].

$$s \longrightarrow t \longrightarrow s' \qquad s \circlearrowright t \qquad s \circlearrowright u \circlearrowright t \qquad 0 \longrightarrow 1 \longrightarrow 2 \longrightarrow \cdots$$

Remark 4.5. A finite acyclic G has a finite path category, and [12] gives preservation for all pairs of topologies on a finite category; Theorem 4.3 locates the loops that break it. In the terms of [1], a set of cut points is a set meeting every cycle; the theorem concerns two such sets and asks that their common part meet every closed walk carrying points of both; for one set, meeting every simple cycle and meeting every closed walk are the same condition, for the pair they are not (Example 4.4(b′)). The bound $\min(|S \setminus T|, |T \setminus S|)$ on the alternation is the number of mutual reconstructions, from the values at S and from the values at T, after which the reconstruction from $S \cap T$ is reached.

Remark 4.6. A presheaf on $\mathcal{P}(G)$ restricts against the direction of the edges, $X(c) \to X(d)$ for a path $d \to c$, so the flows of Section 4 are the backward ones; forward flows are presheaves on $\mathcal{P}(G^{\mathrm{op}})$, the graph with the edges reversed. Closed walks and strongly connected components are the same for G and G^{op}, so the criterion of Theorem 4.3 and the bound are the same in both directions.

Question 4.7. Is $\mathrm{Sh}(J_S) \cap \mathrm{Sh}(J_T) = \mathrm{Sh}(J_{S\cap T})$ for all S, T, i.e. $J_S \vee J_T = J_{S\cap T}$ also when the alternation is unbounded? The intersection of the ideals $(D_S \circ D_T)^k$ is $D_{S\cap T}$ in all cases, and for the infinite path with S the even and T the odd vertices the join is degenerate [11], which is $J_\emptyset$.

5. Finite action monoids

A deterministic transition system with action alphabet Σ is a coalgebra $X \to X^\Sigma$, i.e. a right Σ^*-set, and the coalgebra homomorphisms are the equivariant maps; the presheaf topos of right M-sets for a monoid M is the general form. Its *ideal topologies* are those with a least covering sieve, and they correspond to the idempotent two-sided ideals $D \subseteq M$, with closure $c_D(A) = \{x : x \cdot D \subseteq A\}$ [13, 14]: a state is in the closure of the invariant A iff all its D-futures lie in A.

Lemma 5.1. *In a finite monoid M a two-sided ideal D is idempotent iff it is generated by idempotents, $D = M\,E\,M$ with $E = \{e \in D : e^2 = e\}$.*

Proof. $(MeM)(MeM) \ni ue\,ev = uev$ gives $MEM \subseteq (MEM)^2 \subseteq MEM$. Conversely let $D = D^2$, so $D = D^n$ for all n, and write $d \in D$ as $d = s_1 \cdots s_n$ with $s_i \in D$ and $n > |M|$. Two prefixes coincide, $s_1 \cdots s_i = s_1 \cdots s_j$ with $i < j$, so with $w = s_{i+1} \cdots s_j \in D$ the prefix $p = s_1 \cdots s_i$ satisfies $p = pw^m$ for all m; some power $e = w^m$ is idempotent, $e \in D$, and $d = pe\,s_{j+1} \cdots s_n \in MeM$. □

Proposition 5.2. *On the right M-sets of a finite monoid M the ideal topologies are indexed by the ideals generated by sets E of idempotents, with $c_{MEM}(A) = \{x : x\,u\,e \in A \text{ for all } u \in M, e \in E\}$: a state is in the closure of the invariant A iff every future that passes through a reset $e \in E$ lands in A. By* [13]*, $W_D \vee W_{D'} = W_{J_D \vee J_{D'}}$ for any two of them, with the stabilisation rank of $(DD')^k$ as the bound on the alternation. For the free monoid Σ^*, and more generally for a monoid with a length function additive and positive on non-identity elements, the only idempotent ideals are $\emptyset$ and M, so the deterministic systems over a free alphabet have no non-trivial ideal topology.*

Proof. The first statement is Lemma 5.1 and the description of c_D, as $x \cdot MeM \subseteq A$ iff $x\,u\,e \in A$ for all u (the right factor preserves A). The second is [13] together with finiteness. For the last, a shortest non-identity element of an idempotent ideal $D \neq M$ is a product of two elements of D of larger length. □

Remark 5.3. Nontrivial ideal topologies on deterministic systems thus need relations among the actions, typically idempotent actions (resets, normalisation steps). In a band ($w^2 = w$ for all w) every ideal is idempotent, and for the free band on Σ the ideals D_a of words containing the letter a satisfy $D_a D_b = D_a \cap D_b$: one alternation. The contracted cycle monoids of [12], with stabilisation rank growing with the cycle, are the instances of unbounded alternation in this setting.

6. Questions

Question 6.1. Which pairs of standard numerical abstract domains (intervals, congruences, octagons, polyhedra, on $\mathcal{P}(\mathbb{Z}^n)$) have unbounded alternation at some element, so that the join of the two classes of collapsed pairs is not the class of the least common abstraction? The pairs checked (intervals and congruences) stabilise in two steps because the interval hull of an infinite congruence class is a product of lines and points.

Question 6.2. Extend Theorem 4.3 to flows with transfer functions, i.e. to presheaves on a quotient of the path category by relations among the edges (idempotent edges, commuting edges), where the ideals D_S are replaced by their images; for which relations does the cycle criterion survive?

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of the repository `https://github.com/r-shibusawa/cubical-strict-j`, archived on Zenodo under the concept DOI 10.5281/zenodo.21405961.

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